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existing-structure.json uses the real GS1 Digital Link structure as the existing information structure. It can identify a product through fields such as `gtin`, `batch_or_lot_number`, and `expiration_date`, and it can be accessed through a QR code or 2D barcode. However, it mainly supports product identification and product information access, not a full food safety record.

improved-structure.json is our team's redesigned food safety QR-code structure. It keeps the useful identifier fields from GS1 Digital Link, such as `gtin`, `batch_id`, and `expiration_date`, but adds food safety variables such as `origin_location`, `processing_facility`, `processing_steps`, `inspection_status`, `recall_status`, `verification_status`, and `backend_access_control`. This makes the structure more useful for measuring data quality and managing information security.

How will your structure promote maintaining and measuring quality easier?

Our project focuses on improving food safety information access through a QR-code-based structure that connects food products with origin, processing, inspection, expiration, allergen, and recall information. The improved QR-code food safety structure promotes quality by standardizing food safety information into a consistent JSON format and by providing specific data scopes for tracking product identity, supply chain history, inspection status, allergen information, and recall status.

The improved information structure we did includes specific variables such as `product_id`, `product_name`, `origin_location`, `processing_facility`, `inspection_status`, `inspection_date`, `expiration_date`, and `verification_status`. We think these variables make it easier to measure quality because each food product or batch can be checked for completeness and reliability. In a real world condition, if a record has a `batch_id` but is missing `inspection_date`, `processing_facility`, or `last_recall_check`, the system can flag it as incomplete or outdated. The `verification_status` field also helps measure whether the information has been confirmed by a trusted supplier or regulator.

By transitioning from scattered product labels, supplier spreadsheets, and separate recall webpages to a hierarchical JSON structure, the design creates a shared vocabulary for food safety information. The format can also enforce consistent data types and controlled values. If `inspection_status` can use values such as "passed", "failed", or "pending", while `recall_status` can use values such as "none", "active", or "resolved". This makes

automated quality checks easier because the system can identify missing fields, invalid dates, inconsistent status values, or outdated recall checks.

The design of the improved information structure establishes a single source of truth for each food product or batch. Because the product is identified by stable fields such as `product_id`, `batch_id`, `gtin`, and `qr_code_id`, the system can connect consumer-facing QR-code information with backend supplier records, inspection records, and recall data. Consumers access the current verified record by scanning the QR code, while suppliers and regulators can update or verify the backend data. This makes quality easier to maintain over time because updates to `inspection_status`, `expiration_date`, or `recall_status` can be reflected in one centralized record instead of being scattered across different labels or websites.

How will your structure enable any information security issues be identified and managed down the road?

Our improved structure helps identify and manage information security issues in the future by distinguishing between public consumer information and restricted backend information and adopting role-based access control.

In the existing GS1 Digital Link structure, the main access method is to scan a QR code or two-dimensional barcode to enter the product-related web page information. This enhances the information access capability, but it does not clearly define who can view, update or verify different parts of the record. Our improved structure adds security-related fields such as `consumer_access_view`, `backend_access_control`, `supplier_access`, `regulator_access`, `access_level`, `last_updated_by`, `last_update_time`, `verification_status` and `change_log_enabled`. These variables can help identify security issues because the system can track who modified the record, when the modification occurred, and whether the update was verified. This is particularly important for sensitive security fields such as `inspection_status`, `recall_status`, `recall_reason`, `risk_level` and `verification_status`, as unauthorized modifications may mislead consumers and pose public health risks.

The JSON format also supports information security through logical grouping. For example, `consumer_access_view` lists the fields that can be publicly displayed, such as `product_name`, `origin_location`, `inspection_status`, `expiration_date` and `recall_status`; while `backend_access_control` stores restricted fields, such as supplier permissions, regulatory authority permissions, update history and modification log settings. In terms of data access, the structure adopts a least privilege model. After consumers scan the QR code, they can only access public food safety information in read-only mode. Suppliers can only update their own product records, while regulatory authorities can verify or update higher-risk fields such as `inspection status` and `recall status`. If important values are modified incorrectly or without

authorization, audit fields such as last_updated_by, last_update_time and change_log_enabled can help track and manage the issue subsequently.